Sharp Extremals and Active-Set Evidence for the Goreinov–Tyrtyshnikov–Zamarashkin Hypothesis in the First Open Case

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Abstract

We study the first open case $(n, k) = (6, 3)$ of the Goreinov–Tyrtyshnikov–Zamarashkin row-selection hypothesis. In projector coordinates the problem asks whether every rank-three orthogonal projector $P \in \mathbb{R}^{6\times 6}$ has a 3×3 principal submatrix with least eigenvalue at least $1/6$. We do not prove the hypothesis. Instead we give a computer-assisted exact analysis of the known equality configurations. First, the bound is tight: there are explicit projectors with

$$\max_{|T|=3} \lambda_{\min}(P_{TT}) = \frac{1}{6}.$$

Second, six weighted series–parallel graphic-matroid extremals and one further out-of-family extremal are sharp nonsmooth local minima of the max-eigenvalue functional on $\mathrm{Gr}(3, 6)$. The sharpness certificates are exact rational or number-field computations after positive diagonal changes of coordinates. They imply that all seven configurations are isolated points of the equality set. We also record a symmetry-reduced active-set census and numerical KKT screens over the active-set orbits; no sub-threshold critical point was found. Finally, for one ten-active equality stratum we give a Groebner/CAD cascade which eliminates its positive-dimensional residual determinant branches and classifies the remaining finite residue in the chart. We also include exact finite-branch computations for the two size-twelve equality strata and a zero-dimensional chart computation for the thirteen-active out-of-family stratum. For the most persistent structured size-six low-active survivor, we also give an exact finite-branch exclusion of the full four-parameter ansatz. As background, we include a self-contained Perron-free proof of the weighted Frobenius inequality behind the known $k = 2$ case. We also record the known volume-sampling relaxation

$$\max_{|I|=k} \sigma_{\min}(A_I)^2 \geq \frac{1}{k(n-k)+1},$$

which gives the concrete bound $F(P) \geq 1/10$ in the case $(6, 3)$. For the first open case we prove a stronger relaxed theorem:

$$F(P) \geq \frac{1 - \sqrt{3/5}}{2} > \frac{1}{9},$$

i.e. the GTZ bound with relaxation factor $\alpha = 1/(6F(P)) \leq 1/(3(1 - \sqrt{3/5})) < 3/2$. The remaining global problem is isolated precisely as the exclusion of critical points with value below $1/6$.

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1 Introduction

Let

$$\mathrm{St}(n,k) = \{A \in \mathbb{R}^{n\times k} : A^T A = I_k\}.$$

The Goreinov–Tyrtyshnikov–Zamarashkin hypothesis predicts that for every $A \in \mathrm{St}(n,k)$ one can choose k rows forming a $k \times k$ submatrix A_I with

$$\sigma_{\min}(A_I) \geq \frac{1}{\sqrt{n}}.$$

The cases $k = 1$ and $k = 2$ are known, and by duality the case $k = n-2$ follows from the case 2. Thus $(n,k) = (6,3)$ is the first self-dual open case.

For $A \in \mathrm{St}(6,3)$ set

$$P = AA^T, \qquad F(P) = \max_{|T|=3} \lambda_{\min}(P_{TT}).$$

The right action $A \mapsto AQ$, $Q \in O(3)$, does not change P, so the natural domain is the Grassmannian

$$\mathrm{Gr}(3,6) = \{P = P^T = P^2,\ \mathrm{tr}\, P = 3\}.$$

The case $(6,3)$ is the assertion

$$F(P) \geq \frac{1}{6} \qquad \text{for all } P \in \mathrm{Gr}(3,6). \tag{1.1}$$

This paper is deliberately a partial-result paper. It does not prove (1.1). Its purpose is to record exact local information at the boundary of the conjectured inequality and to make the remaining global problem as concrete as possible.

Main contributions

1. We give a self-contained Perron-free proof of a weighted Frobenius inequality for $k = 2$, implying the usual good-pair step in the core case.

2. We record a standard volume-sampling corollary which gives the relaxed bound $F(P) \geq 1/10$ for all $P \in \mathrm{Gr}(3, 6)$.

3. We prove the stronger relaxed bound $F(P) \geq (1 - \sqrt{3/5})/2 > 1/9$ in the case $(6, 3)$. The proof combines a leverage deletion using the known $(5, 3)$ case with a shifted determinant-sum identity on the high-leverage core.

4. We rederive the tight equality projectors in the weighted graphic-matroid family and verify exactly that $F = 1/6$ there.

5. We give exact sharpness certificates at all six projectors in this series–parallel census. In particular the equality projectors are not flat: F grows linearly in every nonzero Grassmann tangent direction.

6. We record a seventh equality projector, with leverage pattern $(5/14)^{\times 3}, (9/14)^{\times 3}$, outside the weighted series–parallel census, and certify its sharpness exactly.

7. We prove the general implication "sharp equality point implies isolated" for the equality set
$$\mathcal{E} = \{P \in \mathrm{Gr}(3, 6) : F(P) = 1/6\}.$$
Thus the seven certified projectors are isolated equality points.

8. We enumerate active-set orbits under row relabelling and run numerical KKT screens over these orbits. These screens found no sub-threshold critical point.

9. For the hardest observed size-six low-active survivor, we identify a four-parameter structured ansatz. Exact characteristic-zero Groebner and FGLM computations reduce its numerically relevant branch to finite lexicographic systems, and exact real-algebraic sign checks exclude all real points on that branch from satisfying the active and inactive threshold inequalities.

10. For the active set of the base extremal $P(S(e, e, e), e, e, e)$, we record an exact CAD-style cascade: component relations plus active positive semidefiniteness reduce the positive-dimensional determinant locus to a finite infeasible endpoint, while the remaining finite open residue is exactly the known 32-point sign family in the chart.

11. For the thirteen-active out-of-family active set, the determinant chart ideal is already zero-dimensional of degree 32, and its lex basis gives exactly the corresponding 32-point sign family.

12. For the two remaining size-twelve active-set orbits in the known equality census, the exact determinant ideals have only the two possible real branch values $u = \pm 1/9$ for the inverse Gram determinant. Since $u = 1/d(Z) > 0$ on the real chart, the positive branch is forced; on that branch both ideals are zero-dimensional of degree 32 and give exactly their known sign families.

Status conventions

Statements called *proved* below are either conventional arguments or exact finite computations with rational/algebraic arithmetic, implemented in the accompanying scripts listed in Section 16. Floating-point searches are labelled numerical evidence and are not used as proofs.

2 Projector formulation and elementary reductions

For $P = AA^T$ write $\ell_i = P_{ii}$ and $c_{ij} = P_{ij}$. If $T \subset [6]$ has cardinality 3, then

$$P_{TT} = A_T A_T^T,$$

so $\lambda_{\min}(P_{TT}) = \sigma_{\min}(A_T)^2$. Hence the original row-selection bound is exactly (1.1).

Proposition 2.1 (Volume-sampling relaxed bound)**.** *Let $1 \le k < n$ and $A \in \mathrm{St}(n,k)$. Then there is a k-row set $I \subset [n]$ such that*

$$\sigma_{\min}(A_I)^2 \ge \frac{1}{k(n-k)+1}.$$

Equivalently, for $P = AA^T$,

$$\max_{|I|=k} \lambda_{\min}(P_{II}) \ge \frac{1}{k(n-k)+1}.$$

In particular, for $(n,k) = (6,3)$ one has the unconditional relaxed bound $F(P) \ge 1/10$.

Proof. Set $X = A^T \in \mathbb{R}^{k\times n}$, so $XX^T = I_k$. Choose a size-k column set S by volume sampling, i.e. with probability proportional to $\det(X_S X_S^T)$. The volume-sampling expectation formula of Dereziński and Warmuth [4, Theorem 4] gives

$$\mathbb{E}\big[(X_S X_S^T)^{-1}\big] \preceq (n-k+1)(XX^T)^{-1} = (n-k+1)I_k,$$

with equality when the volume-sampling distribution has full support. Taking traces, some sampled set S satisfies

$$\mathrm{tr}\big((X_S X_S^T)^{-1}\big) \le k(n-k+1).$$

Let $0 < \mu_1 \le \cdots \le \mu_k \le 1$ be the eigenvalues of $X_S X_S^T$. The upper bound $\mu_j \le 1$ follows from $X_S X_S^T \preceq XX^T = I_k$. Therefore $\mu_j^{-1} \ge 1$ for $j \ge 2$, and

$$\frac{1}{\sigma_{\min}(X_S)^2} = \mu_1^{-1} \le k(n-k+1) - (k-1) = k(n-k)+1.$$

Since $X_S = A_S^T$, the singular values are the same as those of A_S. □

Lemma 2.2 (Case-A reduction)**.** *Let $1 \le k < n$, and let $A \in \mathrm{St}(n,k)$. If some row has leverage $\ell_i \le 1/n$, then the validity of* GTZ$(n-1,k)$ *implies the desired GTZ conclusion for this particular matrix A.*

Proof. Rotate columns so that row i is $(b,0,\dots,0)$ with $b^2 = \ell_i$. Delete that row and multiply the first column by $(1-\ell_i)^{-1/2}$. The resulting matrix $\widetilde{B}$ lies in $\mathrm{St}(n-1,k)$. If J is a k-row subset of $\widetilde{B}$ and $\varphi(J)$ is the corresponding subset of the original matrix, then

$$A_{\varphi(J)} = \widetilde{B}_J \,\mathrm{diag}\big(\sqrt{1-\ell_i}, 1, \dots, 1\big) Q^T$$

for an orthogonal Q. Therefore

$$\sigma_{\min}(A_{\varphi(J)}) \ge \sqrt{1-\ell_i}\,\sigma_{\min}(\widetilde{B}_J).$$

If $\sigma_{\min}(\widetilde{B}_J) \ge 1/\sqrt{n-1}$, then

$$\sigma_{\min}(A_{\varphi(J)}) \ge \sqrt{1-\frac{1}{n}}\,\frac{1}{\sqrt{n-1}} = \frac{1}{\sqrt{n}}.$$

□

Remark 2.3. Lemma 2.2 does not imply that proving $(6, 3)$ proves all $(n, 3)$. It only removes rows of leverage at most $1/n$. To prove all $(n, 3)$ by induction, one must still prove the core case, where every leverage is $> 1/n$, at every intermediate size.

Theorem 2.4 (A relaxed determinant-sum bound)**.** *For every $P \in \mathrm{Gr}(3, 6)$,*

$$F(P) = \max_{|T|=3} \lambda_{\min}(P_{TT}) \geq t_* := \frac{1 - \sqrt{3/5}}{2}.$$

Proof. Write $P = AA^T$ with $A \in \mathrm{St}(6, 3)$, and let $\ell_i = P_{ii}$. We first prove $F(P) > t$ for every rational $t < t_*$; the stated closed bound then follows by letting $t \uparrow t_*$.

First suppose some leverage satisfies $\ell_i \leq 1 - 5t$. Repeat the deletion and column-rescaling argument from Lemma 2.2. The resulting matrix lies in $\mathrm{St}(5, 3)$. The case $(5, 3)$ is known by duality from the $k = 2$ theorem, so there are three rows of the rescaled matrix with squared least singular value at least $1/5$. Scaling back to the original matrix gives

$$\sigma_{\min}(A_J)^2 \geq \frac{1 - \ell_i}{5} \geq t.$$

It remains to treat the high-leverage core, where $\ell_i > 1 - 5t$ for all i. Then every triple T has

$$\mathrm{tr}(P_{TT}) = \sum_{i \in T} \ell_i > 3(1 - 5t).$$

Let the eigenvalues of P_{TT} be $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq 1$. If $\lambda_2 \leq t$, then

$$\mathrm{tr}(P_{TT}) \leq 1 + 2t < 3(1 - 5t),$$

a contradiction, because $t < t_* < 2/17$. Hence $\lambda_2 > t$ for every triple. This means

$$\lambda_{\min}(P_{TT}) \leq t \iff \det(P_{TT} - tI_3) \leq 0$$

throughout the high-leverage core.

Assume for contradiction that every triple is bad at level t. Then

$$\det(P_{TT} - tI_3) \leq 0 \qquad \text{for all } |T| = 3.$$

On the other hand, summing the characteristic-polynomial coefficients over all triples gives a positive number. Indeed, if $e_r(P_{TT})$ denotes the r-th elementary symmetric polynomial of the eigenvalues of P_{TT}, then

$$\det(P_{TT} - tI_3) = e_3(P_{TT}) - te_2(P_{TT}) + t^2 e_1(P_{TT}) - t^3.$$

Since P is a rank-three projector,

$$\sum_{|T|=3} e_3(P_{TT}) = 1, \qquad \sum_{|T|=3} e_2(P_{TT}) = 4\binom{3}{2} = 12, \qquad \sum_{|T|=3} e_1(P_{TT}) = \binom{5}{2} \mathrm{tr}\, P = 30.$$

Therefore

$$\sum_{|T|=3} \det(P_{TT} - tI_3) = 1 - 12t + 30t^2 - 20t^3.$$

For $t < t_* = (1 - \sqrt{3/5})/2$, the cubic on the right is positive. Indeed the cubic has roots

$$\frac{1 - \sqrt{3/5}}{2}, \qquad \frac{1}{2}, \qquad \frac{1 + \sqrt{3/5}}{2},$$

and value 1 at $t = 0$. Thus

$$\sum_{|T|=3} \det(P_{TT} - tI_3) > 0,$$

contradicting the nonpositivity of every summand. Thus at least one triple has $\lambda_{\min}(P_{TT}) > t$ in the high-leverage core, and the theorem follows. $\square$

3 A Perron-free Frobenius proof for the case $k = 2$

The case $k = 2$ is known by the theorem of Sengupta–Pautov. We record here an independent proof of the weighted pair inequality that gives the good-pair step in the core case. The proof is included because it is short, uses only Frobenius identities and one Rayleigh bound, and clarifies which part of the $k = 2$ argument does not presently extend to triples.

Let $A = [a\ b] \in \mathrm{St}(n, 2)$, and write the i-th row as

$$r_i = (a_i, b_i), \qquad \ell_i = \|r_i\|^2, \qquad c_{ij} = \langle r_i, r_j\rangle, \qquad p_{ij} = a_i b_j - b_i a_j.$$

Thus $\sum_i \ell_i = 2$, $\sum_{i<j} p_{ij}^2 = 1$, and

$$\ell_i \ell_j = c_{ij}^2 + p_{ij}^2.$$

Define the pair bracket

$$b_{ij} = p_{ij}^2 - \frac{\ell_i + \ell_j}{n} + \frac{1}{n^2} = \left(\ell_i - \frac{1}{n}\right)\left(\ell_j - \frac{1}{n}\right) - c_{ij}^2.$$

Equivalently, b_{ij} is the characteristic polynomial of the pair block $P_{\{i,j\}\{i,j\}}$ evaluated at $1/n$.

Theorem 3.1 (Weighted Frobenius inequality)**.** *For every $A \in \mathrm{St}(n, 2)$,*

$$\sum_{i<j} p_{ij}^2 b_{ij} \geq 0.$$

Proof. Set $z_i = a_i + \mathrm{i} b_i$ and $\zeta_i = z_i^2$. Since the two columns of A are orthonormal,

$$\sum_i \zeta_i = 0, \qquad \sum_i |\zeta_i| = 2.$$

Let $w_i = (\operatorname{Re}\zeta_i, \operatorname{Im}\zeta_i)$, let W be the $n \times 2$ matrix with rows w_i, and put

$$K = WW^T, \qquad L = \ell\ell^T, \qquad Q = \sum_i \ell_i^2.$$

Then $K\mathbf{1} = 0$, and

$$K_{ij} = c_{ij}^2 - p_{ij}^2, \qquad L_{ij} = c_{ij}^2 + p_{ij}^2.$$

Hence $p_{ij}^2 = \frac{1}{2}(L - K)_{ij}$ off the diagonal, and the diagonal entries of $L - K$ vanish. Therefore

$$\sum_{i<j} p_{ij}^4 = \frac{1}{8}\|L - K\|_F^2.$$

Using $\sum_{j \neq i} p_{ij}^2 = \ell_i$, one also has

$$\sum_{i<j} p_{ij}^2 b_{ij} = \sum_{i<j} p_{ij}^4 - \frac{Q}{n} + \frac{1}{n^2} = \frac{1}{8}\|L - K\|_F^2 - \frac{Q}{n} + \frac{1}{n^2}.$$

It remains to show $\|L - K\|_F^2 \geq 8Q/n - 8/n^2$. Let $\mu_1 \geq \mu_2 \geq 0$ be the two eigenvalues of $W^T W$. Then $\mu_1 + \mu_2 = Q$ and

$$\|K\|_F^2 = \mu_1^2 + \mu_2^2, \qquad \langle L, K\rangle = \ell^T K \ell.$$

With $v = \ell - \frac{2}{n}\mathbf{1}$, the identity $K\mathbf{1} = 0$ gives $\ell^T K \ell = v^T K v$, while $\|v\|^2 = Q - 4/n$. The Rayleigh bound for the positive semidefinite matrix K gives

$$v^T K v \leq \mu_1 \|v\|^2 = \mu_1 \left(Q - \frac{4}{n}\right).$$

Consequently

$$\begin{aligned}\|L-K\|_F^2-\frac{8Q}{n}+\frac{8}{n^2}&=Q^2-2\ell^TK\ell+\mu_1^2+\mu_2^2-\frac{8Q}{n}+\frac{8}{n^2}\\&\geq Q^2-2\mu_1\left(Q-\frac{4}{n}\right)+\mu_1^2+\mu_2^2-\frac{8Q}{n}+\frac{8}{n^2}\\&=2\left(\mu_2-\frac{2}{n}\right)^2\geq 0,\end{aligned}$$

where the last equality uses $\mu_1=Q-\mu_2$. This proves the claim. □

Corollary 3.2 (Good pair in the core case)**.** *For every $A\in\mathrm{St}(n,2)$ there is a nonsingular pair $i<j$ with*

$$b_{ij}\geq 0.$$

If in addition A is in the core region $\ell_i>1/n$ for all i, then this pair satisfies

$$\lambda_{\min}\left(P_{\{i,j\}\{i,j\}}\right)\geq\frac{1}{n}.$$

Proof. The weights p_{ij}^2 are nonnegative and sum to 1. If every nonsingular pair had $b_{ij}<0$, then the weighted sum in Theorem 3.1 would be negative, a contradiction.

For the final assertion, let $\lambda_1\leq\lambda_2$ be the two eigenvalues of $P_{\{i,j\}\{i,j\}}$. Since b_{ij} is its characteristic polynomial at $1/n$, the inequality $b_{ij}\geq 0$ says that $1/n$ is not strictly between λ_1 and λ_2. In the core region $\lambda_1+\lambda_2=\ell_i+\ell_j>2/n$. Therefore $1/n$ cannot lie above both eigenvalues, and hence $\lambda_1\geq 1/n$. □

Remark 3.3. Theorem 3.1 is a pair theorem. It proves the key core-case step for $k=2$, and the usual Case-A induction supplies the boundary rows. It does not by itself produce a good triple when $k=3$. In fact, the natural volume-weighted analogues of this averaged inequality fail in higher rank, so the $(6,3)$ problem requires the separate active-set and semialgebraic analysis developed below.

4 The nonsmooth active-set problem

The functional

$$F(P)=\max_{|T|=3}g_T(P),\qquad g_T(P)=\lambda_{\min}(P_{TT}),$$

is a maximum of twenty eigenvalue functions. At a point where all active least eigenvalues are simple, g_T is differentiable for active T and

$$dg_T(P)[\dot{P}]=v_T^T\dot{P}_{TT}v_T,$$

where v_T is a unit least-eigenvector of P_{TT}.

Definition 4.1. The active set at P is

$$\mathcal{A}(P)=\{T:g_T(P)=F(P)\}.$$

If the active least eigenvalues are simple, we call P Clarke-stationary for F if

$$0\in\mathrm{conv}\{dg_T(P):T\in\mathcal{A}(P)\}$$

after restriction to the tangent space $T_P\,\mathrm{Gr}(3,6)$.

Proposition 4.2 (KKT reduction). *The inequality* (1.1) *holds if and only if no Clarke-stationary point of* F *on* $\mathrm{Gr}(3,6)$ *has value below* $1/6$.

Proof. If a sub-threshold point exists, F attains a sub-threshold global minimum on the compact manifold $\mathrm{Gr}(3,6)$, and such a minimizer is Clarke-stationary. The converse is immediate. □

Thus the whole remaining global problem can be phrased as:

$$\text{exclude all critical points of } F \text{ with } F < 1/6.$$

5 The weighted graphic-matroid equality projectors

We recall the construction used for the six series–parallel equality projectors. Let G be a connected graph with six edges, choose a reduced incidence matrix B_{red}, and let $W = \mathrm{diag}(w_1, \ldots, w_6)$ with $w_i > 0$. Define

$$\widetilde{M} = W^{1/2} B_{\text{red}}^T, \qquad P = \widetilde{M}(\widetilde{M}^T \widetilde{M})^{-1} \widetilde{M}^T.$$

Then P is a rank-three orthogonal projector when G has four vertices. The leverage of an edge is

$$\ell_e = w_e R_{\text{eff}}(e),$$

where $R_{\text{eff}}(e)$ is the effective resistance between the endpoints of the edge in the weighted network.

The exact census in the accompanying script enumerates the two-terminal series–parallel trees on six edges and solves for weights with $F = 1/6$. It finds the following six configurations.

Graph expression	weights	leverages	$\lvert\mathcal{A}\rvert$
$P(S(e,e,e),e,e,e)$	$1,1,1,\frac{5}{9},\frac{5}{9},\frac{5}{9}$	$(\frac{13}{18})^3,(\frac{5}{18})^3$	10
$P(S(e,e),S(e,e),e,e)$	$1,1,1,1,\frac{5}{8},\frac{5}{8}$	$(\frac{11}{18})^4,(\frac{5}{18})^2$	12
$P(S(P(e,e,e),e),S(e,e))$	$1,1,1,\frac{9}{5},\frac{9}{5},\frac{9}{5}$	$(\frac{5}{18})^3,(\frac{13}{18})^3$	10
$P(S(P(S(e,e),e,e),e),e)$	$1,1,\frac{5}{8},\frac{5}{8},1,1$	$(\frac{11}{18})^4,(\frac{5}{18})^2$	12
$P(S(P(e,e),P(e,e)),S(e,e))$	$1,1,1,1,\frac{8}{5},\frac{8}{5}$	$(\frac{7}{18})^4,(\frac{13}{18})^2$	12
$P(S(P(e,e),e),S(P(e,e),e))$	$1,1,\frac{8}{5},1,1,\frac{8}{5}$	$(\frac{7}{18})^4,(\frac{13}{18})^2$	12

Proposition 5.1. *For each of the six projectors above,* $P^2 = P$, $\operatorname{tr} P = 3$, *and*

$$F(P) = \frac{1}{6}.$$

Moreover every active 3×3 *block has simple least eigenvalue.*

Proof. The statements are finite exact checks. The projector identities are rational. For each triple T, the characteristic polynomial of P_{TT} is computed exactly; the active triples have least eigenvalue $1/6$, while the inactive triples in this census are singular with least eigenvalue 0. Simplicity is checked by verifying that the second eigenvalue of each active block is strictly larger than $1/6$. □

6 Sharpness certificates

Let $P_0 \in \mathcal{E}$, where

$$\mathcal{E} = \{P \in \mathrm{Gr}(3,6) : F(P) = 1/6\}.$$

Assume that the active least eigenvalues at P_0 are simple. Choose a basis $B_1, \dots, B_9$ of $T_{P_0}\,\mathrm{Gr}(3,6)$. For each active triple T, let v_T be a least-eigenvector of $(P_0)_{TT}$ and set

$$L_{Tj} = v_T^T (B_j)_{TT} v_T.$$

Then for tangent coordinates $z \in \mathbb{R}^9$,

$$F'(P_0; z) = \max_{T \in \mathcal{A}(P_0)} (Lz)_T.$$

Definition 6.1. We call P_0 *sharp* if

$$\kappa(P_0) = \min_{\|z\|=1} \max_{T \in \mathcal{A}(P_0)} (Lz)_T > 0.$$

Lemma 6.2 (Sharpness criterion)**.** *Suppose* $\operatorname{rank} L = 9$ *and there is a vector* λ *with all entries strictly positive such that*

$$L^T \lambda = 0.$$

Then P_0 *is sharp.*

Proof. If $\max_T (Lz)_T \le 0$, then

$$0 = \lambda^T L z = \sum_T \lambda_T (Lz)_T.$$

This is a sum of nonpositive terms with strictly positive coefficients, hence every $(Lz)_T = 0$. Since $\operatorname{rank} L = 9$, this forces $z = 0$. Thus $\max_T (Lz)_T > 0$ for all $z \ne 0$, and compactness of the unit sphere gives a positive minimum. □

For some of the projectors the positive multiplier is not the most convenient certificate. The following equivalent rational LP certificate was used as the load-bearing test. Maximize

$$\sum_T (-Lz)_T$$

over the rational polytope

$$Lz \le 0, \qquad \|z\|_\infty \le 1.$$

The feasible point $z = 0$ has objective 0, and all summands are nonnegative on the feasible set. If the exact optimum is 0 and $\operatorname{rank} L = 9$, then every feasible z satisfies $Lz = 0$, hence $z = 0$. Since $Lz \le 0$ is a cone, this proves that the critical cone is trivial and $\kappa(P_0) > 0$.

Theorem 6.3 (Sharpness of the six graphic-matroid extremals)**.** *All six projectors in Proposition 5.1 are sharp points of* F *on* $\mathrm{Gr}(3,6)$.

Proof. The proof is by exact rational computation. A direct orthonormal tangent basis contains square roots. These are eliminated as follows. Put

$$\widetilde{B} = B_{\mathrm{red}}^T, \qquad G = \widetilde{B}^T W \widetilde{B}, \qquad S = \widetilde{B}\, G^{-1} \widetilde{B}^T W.$$

Then S is a rational idempotent and

$$P = W^{1/2} S W^{-1/2}.$$

Because $W^{1/2}$ is diagonal, this similarity restricts to every principal block, so P_{TT} and S_{TT} have the same spectrum. The active eigenvectors, tangent functionals, and the LP certificate can therefore be written over $\mathbb{Q}$ after positive row scalings of L. For all six cases the exact computation gives $\operatorname{rank} L = 9$ and exact LP optimum 0. □

7 A seventh extremal outside the graphic census

Numerical descent from random starts repeatedly finds an equality point whose sorted leverage pattern is

$$\left(\frac{5}{14}, \frac{5}{14}, \frac{5}{14}, \frac{9}{14}, \frac{9}{14}, \frac{9}{14}\right).$$

This pattern is absent from the six graphic-matroid projectors above. The projector was reconstructed exactly over $\mathbb{Q}(\sqrt{2}, \sqrt{5})$. Its off-diagonal entries have magnitudes among

$$\frac{5}{14}, \qquad \frac{\sqrt{5}}{21}, \qquad \frac{5\sqrt{5}}{42}, \qquad \frac{5\sqrt{2}}{42}, \qquad \frac{\sqrt{10}}{14}.$$

Theorem 7.1 (The out-of-family extremal). *There is an exact projector $P \in \mathrm{Gr}(3,6)$ with leverage pattern $(5/14)^{\times 3}, (9/14)^{\times 3}$ such that*

$$F(P) = \frac{1}{6}.$$

It has thirteen active triples and seven singular triples. Every active least eigenvalue is simple, and P is a sharp point of F on $\mathrm{Gr}(3,6)$.

Proof. The entries are reconstructed from the numerical projector by PSLQ and then checked exactly in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$. The projector identities $P^T = P$, $P^2 = P$, $\operatorname{tr} P = 3$, the leverage pattern, and the active/singular triple counts are exact finite checks. The sharpness certificate uses the same critical-cone LP as Theorem 6.3. In this case an additional positive column scaling of the tangent basis rationalizes L; positive diagonal coordinate changes preserve both $\operatorname{rank} L$ and the property

$$\{z : Lz \le 0\} = \{0\}.$$

The resulting rational matrix has rank 9 and exact LP optimum 0. □

Remark 7.2. An exhaustive search over the two-terminal series–parallel trees with the weight parametrization used for the six graphic projectors does not produce the leverage pattern $(5/14)^{\times 3}, (9/14)^{\times 3}$. Thus the seventh point is not merely a relabelling of the displayed graphic census.

8 Isolation and conditional finiteness

Lemma 8.1 (Sharp points are isolated). *Let $P_0 \in \mathcal{E}$ be sharp and assume the active least eigenvalues at P_0 are simple. Then there is a neighborhood U of P_0 in $\mathrm{Gr}(3,6)$ such that*

$$U \cap \mathcal{E} = \{P_0\}.$$

Proof. Near P_0, inactive triples remain below $1/6$ by continuity. Since the active least eigenvalues are simple, the active functions g_T are analytic. Hence, for $P = P_0 + r\dot{P}$ in a local chart and $\|\dot{P}\| = 1$,

$$F(P) = \frac{1}{6} + rF'(P_0; \dot{P}) + O(r^2)$$

uniformly in $\dot{P}$. Sharpness gives $F'(P_0; \dot{P}) \ge \kappa > 0$ on the unit sphere, so $F(P) > 1/6$ for sufficiently small $r > 0$. □

Corollary 8.2. *The seven projectors in Theorems 6.3 and 7.1 are isolated points of $\mathcal{E}$.*

Theorem 8.3 (Conditional finiteness). *If every point of $\mathcal{E}$ is sharp, then $\mathcal{E}$ is finite.*

Proof. The set $\mathcal{E} = F^{-1}(1/6)$ is closed in the compact manifold $\mathrm{Gr}(3,6)$, hence compact. If every point is sharp, Lemma 8.1 makes $\mathcal{E}$ a compact discrete space. A compact discrete space is finite. □

Remark 8.4. The hypothesis of Theorem 8.3 is not proved. It is a global statement over $\mathcal{E}$, and sampling cannot discharge it. The theorem is nevertheless useful because it identifies a concrete local certificate whose failure would point to a positive-dimensional equality family.

9 Active-set orbit census and numerical KKT screens

The twenty triples of [6] are acted on by the row-permutation group S_6. Burnside's lemma gives the following numbers of active-subset orbits by cardinality:

$\lvert\mathcal{A}\rvert$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
#	1	1	3	7	21	43	94	161	249	312	352	312	249	161	94	43	21	7	3	1	1.

Thus there are 2136 active-set orbits total, 579 nonempty orbits with $\lvert\mathcal{A}\rvert \le 8$, and 1556 orbits with $\lvert\mathcal{A}\rvert \ge 9$. The seven certified equality projectors occupy four distinct active-set orbits.

For a selected active set $\mathcal{A}_0$, the numerical screen minimizes residuals for the following conditions:

$$g_T(P) = t \quad (T \in \mathcal{A}_0), \qquad g_T(P) \le t \quad (T \notin \mathcal{A}_0),$$

together with the stationarity condition

$$0 = \Pi_{T_P\,\mathrm{Gr}(3,6)}\left(\sum_{T\in\mathcal{A}_0} \alpha_T\, dg_T(P)\right), \qquad \alpha_T \ge 0, \quad \sum_T \alpha_T = 1.$$

For counterexample searches the level condition is $t < 1/6$; for equality searches it is $t = 1/6$. This is a heuristic least-squares screen, not a proof.

Observed numerical results

mode	active sizes	orbit reps	candidates	note
counterexample	$1,\dots,4$	32	0	no tripwire
equality	$1,\dots,4$	32	0	no low-active point
counterexample	$5,6$	137	0	no tripwire
equality	$5,6$	137	0	no low-active point
counterexample	$7,8$	410	0	no tripwire
equality	$7,8$	410	1	selected subset of known 12-active point
counterexample	9	312	0	one near-threshold non-candidate
equality	9	312	0	no equality stratum
counterexample	$10,\dots,13$	1074	0	MLCore run
equality	$10,\dots,13$	1074	2	selected subsets of known 12- and 13-active points
counterexample	$14,\dots,20$	170	0	no tripwire
equality	$14,\dots,20$	170	0	no equality stratum

The equality hit with selected $\lvert\mathcal{A}_0\rvert = 8$ had residual about $7.1 \cdot 10^{-13}$, but its actual active set had size 12 and matched a known TTSP active-set orbit. Thus it is not an eight-active equality stratum.

The two equality hits from selected sizes $10,\dots,13$ were likewise selected-subset hits. Reconstructing the projectors from the numerical parameters and recomputing the actual active sets gave one known TTSP 12-active orbit and the out-of-family 13-active orbit. In both cases the actual active-gradient matrix has rank 9 and a strictly positive numerical multiplier, so neither is a non-sharp candidate.

The best size-nine counterexample near-miss polished to

$$F = 0.16666994562575838 > 1/6$$

with actual top active size one. The best MLCore record for selected $|\mathcal{A}_0| = 12$ had

$$F = 0.166677230 > 1/6$$

and again actual top active size one. No numerical run found a point with $F < 1/6 - 10^{-6}$.

Across all equality-mode screens, the actual-active-set classifier found three low-residual equality hits, all in known sharp active-set orbits; it found no new actual active-set orbit, no actual active size at most 9, and no numerical non-sharp signature.

Remark 9.1. These screens are useful mainly as triage. They also expose a bookkeeping hazard: a selected subset of a larger actual active set may satisfy equality and stationarity to high precision. Any exact active-set proof must therefore track the actual active set, not only the selected equations.

10 A semialgebraic finiteness target

The conditional finiteness theorem reduces finiteness of $\mathcal{E}$ to excluding non-sharp equality points. In the simple-active case this obstruction has a small polynomial representation. Work in the Grassmann chart

$$Y = \begin{pmatrix} I_3 \\ Z \end{pmatrix}, \qquad P(Z) = Y(Y^TY)^{-1}Y^T = \frac{N(Z)}{d(Z)}, \qquad d = \det(Y^TY) > 0.$$

For a triple T, put

$$M_T(Z) = 6N(Z)_{TT} - d(Z)I_3.$$

Then $\lambda_{\min}(P(Z)_{TT}) = 1/6$ with simple least eigenvalue is encoded by

$$M_T(Z) \succeq 0, \qquad \det M_T(Z) = 0, \qquad \operatorname{rank} M_T(Z) = 2.$$

The raw determinant has the universal factor d^2; since this chart has $d > 0$, the active determinant equation may equivalently be written as the saturated degree-six equation

$$\frac{\det M_T(Z)}{d(Z)^2} = 0.$$

In purely algebraic computations this open chart condition should be retained, for instance by adjoining an inverse variable u and the equation $u\,d(Z) - 1 = 0$. On a cofactor patch, a nonzero kernel vector $w_T(Z)$ is obtained as the cross product of two rows of M_T. If $h \in \mathbb{R}^9$ is a nonzero tangent witness for non-sharpness, then after clearing the positive denominator d^2 one obtains the polynomial inequalities

$$w_T^T \left[\sum_j h_j \big(d\,\partial_j N - N\,\partial_j d \big)_{TT} \right] w_T \le 0, \qquad T \in \mathcal{A},$$

with the normalization $\sum_j h_j^2 = 1$.

Thus the simple-active non-sharp obstruction lives in only 18 variables: nine chart variables and nine tangent variables. The active-set size changes the number of equations and inequalities, but not the ambient dimension. A degree audit gives $\deg d = 6$, entries of N of degree at most 6, raw active determinant degree at most 18 but saturated active determinant degree 6, cofactor-patch norm degree at most 24, and cleared directional inequalities of degree at most 35.

There is also a useful count. At a simple-active equality point, sharpness means that the active gradients positively span the 9-dimensional tangent space. This requires at least ten active gradients. Hence a simple-active equality point with $|\mathcal{A}| \leq 9$ would automatically be non-sharp. The equality screens found no such point, and all observed equality points have $|\mathcal{A}| \in \{10, 12, 13\}$.

The nonsimple case, where an active least eigenvalue has multiplicity greater than one, must be split off separately: one active triple then contributes a set-valued subgradient rather than one row of the active-gradient matrix.

11 A CAD cascade for the base active set

The semialgebraic target becomes more concrete on the active set of the base graphic extremal $P(S(e,e,e),e,e,e)$. We use zero-based row labels and write

$$\mathcal{A}_0 = \{012, 013, 014, 015, 023, 024, 025, 123, 124, 125\},$$

where, for instance, 013 denotes the triple $\{0,1,3\}$. Let $I_0 \subset \mathbb{Q}[z_0, \ldots, z_8, u]$ be generated by the ten saturated active determinants

$$\frac{\det(6N(Z)_{TT} - d(Z)I_3)}{d(Z)^2}, \qquad T \in \mathcal{A}_0,$$

together with the chart equation $u\,d(Z) - 1 = 0$. For an active triple T and two local row indices $E \subset \{0,1,2\}$, let $\Delta_{T,E}$ denote the corresponding principal 2×2 minor of $6N(Z)_{TT} - d(Z)I_3$.

The following three polynomial relations were found and then checked by open-locus Groebner computations:

$$\begin{aligned} R_0 &= \Delta_{013,\{0,2\}} + 46656 z_0^2, \\ R_1 &= \Delta_{013,\{1,2\}} + 46656 z_1^2, \\ R_2 &= \Delta_{014,\{0,2\}} + 46656 z_7^2. \end{aligned}$$

On a real point satisfying active positive semidefiniteness, each displayed minor is nonnegative. Therefore $R_i = 0$ forces the corresponding square coordinate to vanish.

Proposition 11.1 (Computed cascade on the base stratum)**.** *Over $\mathbb{Q}$ in the full-rank chart $d > 0$, the following statements hold.*

1. *The determinant ideal I_0 has dimension 3. The open locus $I_0 + \langle aR_0 - 1\rangle$ is zero-dimensional of degree 608. Hence R_0 vanishes on every positive-dimensional irreducible component of $V(I_0)$.*

2. *Put $I_1 = I_0 + \langle R_0, z_0\rangle$. Then $\dim I_1 = 2$ and the top degree is 16. Moreover $I_1 + \langle aR_1 - 1\rangle$ is empty, so R_1 holds on this whole branch.*

3. *Put $I_2 = I_1 + \langle R_1, z_1\rangle$. Then $\dim I_2 = 1$ and the top degree is 24. The normal form of R_2 modulo I_2 is zero, so $R_2 \in I_2$.*

4. *Put $I_3 = I_2 + \langle R_2, z_7\rangle$. Then I_3 is zero-dimensional of degree 56, and a lexicographic basis is*

$$z_2^2 - 5, \quad 216u - 1, \quad z_8^2, \quad z_7, \quad z_6^2 - 5, \quad z_5^2,$$
$$z_4^2 - 5, \quad z_3 z_5 z_8, \quad z_3^2, \quad z_1, \quad z_0.$$

Its real radical is therefore

$$z_0 = z_1 = z_3 = z_5 = z_7 = z_8 = 0, \qquad z_2^2 = z_4^2 = z_6^2 = 5, \qquad u = 1/216.$$

The resulting eight real sign choices all fail the GTZ semialgebraic conditions: an active block has $\lambda_{\min} - 1/6 = -1/6$*, and an inactive block has least eigenvalue* $2/3$ *above the threshold.*

5. *The finite open residue from the first step is also accounted for. Let* $J_0 = I_0 + \langle aR_0 - 1\rangle$*. The computed* z_0 *eliminant of* J_0 *is*

$$\begin{aligned} z_0^{10} - \frac{815}{441}z_0^8 - \frac{93850}{64827}z_0^6 + \frac{2295875}{453789}z_0^4 \\ - \frac{13883125}{4084101}z_0^2 + \frac{8556250}{12252303} \\ = (z_0^2 - \tfrac{5}{9})(z_0^2 - \tfrac{10}{21})(z_0^2 + \tfrac{5}{3})\left(z_0^4 - \frac{365}{147}z_0^2 + \frac{34225}{21609}\right). \end{aligned}$$

The last two factors have no real zeros. On the factor $z_0^2 = 5/9$*, an exact lex basis reduces to*

$$9z_1^2 - 5, \quad 25980a - 1, \quad 6u - 1, \quad 9z_8^2 - 5, \quad 9z_7^2 - 5, \quad 9z_6^2 - 5, \quad 9z_5^2 - 5,$$
$$5z_4 - 9z_5z_7z_8, \quad 5z_3 - 9z_5z_6z_8, \quad 5z_2 - 9z_7z_8z_1, \quad 5z_0 - 9z_6z_7z_1.$$

This gives 32 *real sign choices; direct substitution shows that all have the same active set* $\mathcal{A}_0$ *and satisfy the sharp equalities and inequalities. On the factor* $z_0^2 = 10/21$*, the* z_1 *eliminant factors as*

$$(z_1^2 - \tfrac{10}{21})(z_1^2 + \tfrac{5}{3})\left(z_1^4 - \frac{365}{147}z_1^2 + \frac{34225}{21609}\right).$$

The only real subfactor is $z_1^2 = 10/21$*, and adjoining it gives a lex basis whose first polynomial is* $3z_2^2 + 5$*. Hence this branch has no real points.*

Proof. The algebraic dimension and degree statements are Groebner-basis computations using Singular over $\mathbb{Q}$ with its modular backend. The open-locus tests adjoin a fresh inverse variable a and the equation $aR_i - 1 = 0$; if this ideal has smaller dimension, or is empty, then R_i cannot be nonzero on the corresponding component. The positive-semidefinite forcing step is elementary: on the active semialgebraic locus the minors $\Delta_{T,E}$ are nonnegative, and so are the displayed squares. The forced finite lex basis gives the eight real points explicitly, and direct exact substitution checks the active and inactive eigenvalue inequalities. The open-residue eliminant and the displayed real-factor refinements were computed by exact characteristic-zero Groebner computations; a separate six-prime CRT reconstruction gives the same z_0 eliminant. □

Remark 11.2. Proposition 11.1 is not yet a proof of the full $(6, 3)$ case. It accounts for the positive-dimensional residual determinant branches and the finite $R_0 \neq 0$ residue attached to the base active set in the standard chart. The size-twelve and thirteen-active chart computations below propagate the same finiteness program to the other known equality active-set orbits in this standard chart. Other active-set orbits and chart patches remain.

12 A finite chart for the thirteen-active point

The out-of-family extremal of Theorem 7.1 gives another exact chart computation. Let

$$\mathcal{A}_7 = \{012, 014, 015, 023, 024, 025, 034, 035, 123, 124, 125, 134, 135\}.$$

Let I_7 be the saturated determinant ideal for these active triples, again in the standard chart $Y = [I; Z]$ with $u\,d(Z) - 1 = 0$.

Proposition 12.1 (Computed finite chart for the seventh point). *The ideal I_7 is zero-dimensional of degree* 32 *over* $\mathbb{Q}$. *A lexicographic basis is*

$$8z_0^2 - 5, \quad 63u - 8, \quad 8z_8^2 - 5, \quad 64z_7^2 - 45, \quad 64z_6^2 - 45, \quad 8z_5^2 - 5,$$
$$5z_4 - 8z_5 z_7 z_8, \quad 5z_3 - 8z_5 z_6 z_8, \quad z_2, \quad 45z_1 - 64z_6 z_7 z_0.$$

Consequently the real chart points form a 32-point sign family. Direct substitution shows that all 32 *points satisfy the GTZ inequalities with* $F = 1/6$, *and their actual active set is exactly* $\mathcal{A}_7$.

Proof. The displayed lex basis is the output of an exact Singular computation over $\mathbb{Q}$ using the modular backend. It leaves five independent signs, hence 32 real points. The active and inactive inequalities are then checked by substituting the displayed formulas into the 3×3 principal blocks. □

13 Finite real branches for the two size-twelve charts

Two known equality active-set orbits have twelve active triples in the standard chart:

$$\mathcal{A}_{12a} = \{012, 013, 023, 024, 025, 034, 035, 123, 124, 125, 134, 135\},$$
$$\mathcal{A}_{12b} = \{012, 013, 014, 015, 024, 025, 034, 035, 124, 125, 134, 135\}.$$

Let I_{12a} and I_{12b} be the corresponding saturated determinant ideals with $u\,d(Z) - 1 = 0$.

Proposition 13.1 (Computed finite real branches for the size-twelve strata). *For each of* I_{12a} *and* I_{12b}, *an exact Groebner basis contains*

$$729u^3 + 81u^2 - 9u - 1 = 729\left(u - \frac{1}{9}\right)\left(u + \frac{1}{9}\right)^2.$$

Since $d(Z) = \det(Y^T Y) > 0$ *on the real full-rank chart and* $u = 1/d(Z)$, *every real chart point lies on the branch* $u = 1/9$. *On this branch both ideals are zero-dimensional of degree* 32 *over* $\mathbb{Q}$.

For $\mathcal{A}_{12a}$, *a lexicographic basis of the real branch is*

$$z_0^2 - 1, \quad 9u - 1, \quad z_8, \quad 8z_7^2 - 5, \quad 8z_6^2 - 5, \quad z_5, \quad 8z_4^2 - 5,$$
$$5z_3 - 8z_4 z_6 z_7, \quad z_2^2 - 1, \quad 5z_1 - 8z_6 z_7 z_0.$$

For $\mathcal{A}_{12b}$, *a lexicographic basis of the real branch is*

$$z_2^2 - 1, \quad 9u - 1, \quad z_8^2 - 1, \quad 8z_7^2 - 5, \quad 8z_6^2 - 5, \quad z_5^2 - 1,$$
$$z_4 - z_5 z_7 z_8, \quad z_3 - z_5 z_6 z_8, \quad z_1, \quad z_0.$$

Thus each real branch consists of a 32-point sign family. Direct substitution checks that every point in each family satisfies the GTZ inequalities with $F = 1/6$, *and that the actual active set is exactly the selected twelve-triple set.*

Proof. The branch polynomial in u is read from exact characteristic-zero Groebner bases. The displayed lex bases are exact computations after adding $9u - 1$. The sign counts and active/inactive inequalities are verified by substituting the resulting formulas into the projector $P(Z) = Y(Y^T Y)^{-1} Y^T$. □

14 A structured low-active branch

The active-set census also leaves low-active determinant strata that are much harder than the known equality strata. The most persistent size-six survivor in our searches has

$$\mathcal{A}_6 = \{012, 013, 045, 145, 234, 235\}.$$

A refined over-tied numerical root has five outside triples tied at a common level:

$$\{024, 034, 125, 135, 245\}.$$

In the standard chart the root has the pattern

$$Z = \begin{pmatrix} -a & a & -b \\ c & d & -a \\ -d & -c & -a \end{pmatrix}.$$

Substituting this ansatz into the six active determinant equations and five outside-tie determinant equations gives a zero-dimensional system of degree 280, both over $\mathbb{F}_{32003}$ and over $\mathbb{Q}$. The exact $\mathbb{Q}$ grevlex basis contains the simple tail relations

$$a^2((c+d)^2 - 5) = 0, \qquad ((c+d)^2 - 5)((c-d)^2 - 5) = 0.$$

Consequently every real point of the ansatz system lies either on $a = 0$ or on $(c+d)^2 = 5$. The $a = 0$ branch is zero-dimensional of degree 96; an exact FGLM conversion gives a six-polynomial lex basis, and Sage enumeration over the algebraic real field finds no real points. It remains to handle $(c+d)^2 = 5$, which is the branch containing the refined numerical root.

On this branch, an exact $\mathbb{Q}$ grevlex computation gives quotient degree 184, matching the modular computation. Modular lex computations over six primes have stable quotient degree 184 and a degree-22 eliminant in the tie parameter q, where the tied outside triples have least eigenvalue $q/6$. The visible factor

$$(q-3)(q-5)^2(q^2-3q+6)^2(q^2-9q+24)^2$$

removes nuisance branches. The six-prime CRT reconstruction of the residual factor is

$$\begin{aligned} q^{11} &- \frac{575}{16}q^{10} + \frac{562849}{1024}q^9 - \frac{4768527}{1024}q^8 + \frac{97733649}{4096}q^7 \\ &- \frac{311529979}{4096}q^6 + \frac{154321449}{1024}q^5 - \frac{374696325}{2048}q^4 + \frac{543960225}{4096}q^3 \\ &- \frac{225237375}{4096}q^2 + \frac{24046875}{2048}q - \frac{253125}{256}, \end{aligned}$$

which factors over $\mathbb{Q}$ as a cubic times an octic:

$$\begin{aligned} &\left(q^3 - 9q^2 + \frac{81}{4}q - \frac{45}{4}\right) \\ &\cdot \left(q^8 - \frac{431}{16}q^7 + \frac{293857}{1024}q^6 - \frac{776859}{512}q^5 + \frac{2094513}{512}q^4 - \frac{1353145}{256}q^3 \right. \\ &\qquad \left. + \frac{3301125}{1024}q^2 - \frac{453375}{512}q + \frac{5625}{64}\right). \end{aligned}$$

Multiplying this residual factor by the visible nuisance factor and reducing against the exact $\mathbb{Q}$ grevlex basis of the branch gives zero normal form. Thus the degree-22 q-polynomial is certified in characteristic zero; the CRT step is no longer only modular evidence.

Converting the exact branch basis to lexicographic order by FGLM gives quotient dimension 184, again matching the grevlex and modular computations. After adjoining the residual factors separately, the cubic branch has lex quotient dimension 24 and no real points: its lex basis contains $b^2 + 1$ and a^2. The octic branch has lex quotient dimension 32 and contains 16 real points. The nuisance factors $q = 3$ and $q = 5$ have lex quotient dimension 32 each, with 8 and 16 real points respectively.

The refined numerical root corresponds to the real octic root

$$q = 1.180695836766579\ldots,$$

so the tied inactive triples sit above the threshold by

$$\frac{q-1}{6} = 0.030115972794\ldots.$$

Theorem 14.1 (Exclusion of the structured size-six ansatz)**.** *There is no real point in the four-parameter structured ansatz above which realizes the selected threshold pattern: the six selected triples at threshold with $P_{TT} - I/6 \succeq 0$, and every inactive triple satisfying $\lambda_{\min}(P_{TT}) \leq 1/6$.*

Proof. The tail relation $a^2((c+d)^2 - 5) = 0$ splits the ansatz system into the $a = 0$ branch and the $(c+d)^2 = 5$ branch. The $a = 0$ branch has an exact zero-dimensional lex basis of quotient dimension 96, and exact enumeration over the algebraic real field gives no real points.

On the $(c + d)^2 = 5$ branch, the exact $\mathbb{Q}$ branch ideal has quotient dimension 184, and its q-eliminant is the certified degree-22 polynomial above. The non-real quadratic nuisance factors have no real points. The cubic residual factor has no real points because the exact lex basis on that component contains $b^2 + 1$. It remains to inspect the octic residual factor and the real nuisance factors $q = 3$ and $q = 5$.

For these three lex systems, Sage enumeration over the algebraic real field finds 16, 8, and 16 real points respectively. At each point we compute the exact numerator

$$N = Y\ \mathrm{adj}(Y^T Y)\, Y^T, \qquad d = \det(Y^T Y) > 0,$$

so that $P_{TT} - I/6 \succeq 0$ is equivalent to

$$6N_{TT} - dI \succeq 0.$$

For an active triple, a negative principal minor of this shifted block excludes the point. For an inactive triple, a positive-definite shifted block certifies $\lambda_{\min}(P_{TT}) > 1/6$, also excluding a point from the desired threshold pattern.

The exact sign certificate classifies all 40 real points. On the octic component, 12 points have a negative active principal minor and 4 points have an inactive positive-definite shifted block. On the $q = 3$ component, all 8 real points fail active positive semidefiniteness. On the $q = 5$ component, all 16 real points have an inactive positive-definite shifted block. There are no unclassified real points. $\square$

A deterministic numerical classifier using the lifted q-values on the $(c+d)^2 = 5$ branch found the same 40-point real-root profile in two independent runs. This is now only a diagnostic cross-check; the exclusion above is the exact real-algebraic certificate for the structured ansatz.

Remark 14.2. This theorem excludes only the displayed four-parameter structured ansatz for the selected size-six active and outside-tie pattern. It does not prove the full inequality (1.1).

15 What remains

Proposition 4.2 isolates the remaining difficulty: prove that no Clarke-stationary point of F has value below $1/6$. The exact local analysis above, including the CAD cascade in Proposition 11.1 and the finite branch computations in Sections 12 and 13, and the structured low-active evidence in Section 14, does not solve that global problem.

The most realistic continuation is a semialgebraic exclusion of the non-sharp obstruction just described. For each active-set stratum, modulo row-permutation symmetry and cofactor-patch choices, one can ask an exact real-algebraic solver to certify that the equality, positivity, and tangent-witness conditions have no solution outside the known sharp active-set orbits. Inactive constraints may be relaxed for an infeasibility proof: if the relaxed obstruction is empty, then the true obstruction is empty.

A second, more direct but currently less effective, continuation is certified global optimization over $\mathrm{Gr}(3,6)$:

1. Use the exact sharpness certificates to remove explicit neighborhoods of the seven known equality points. A second-order eigenvalue bound gives a radius of the form
$$F(P_0 + r\dot{P}) \geq \frac{1}{6} + \kappa r - \frac{r^2}{\mathrm{gap}_{\min}},$$
where $\mathrm{gap}_{\min}$ is the smallest active spectral gap.

2. Cover the remaining compact set by interval boxes in a Grassmann chart or by several charts, using interval bounds for the twenty 3×3 least eigenvalues.

3. Branch on active triples near nonsmooth regions. The blocks are only 3×3, so interval characteristic-polynomial or Rayleigh quotient bounds are feasible.

This route is computational rather than elegant, but it is honest: it targets the global $\forall$ statement directly and does not rely on extrapolating from sampled extremals.

A relaxed bound $F(P) \geq 1/6 - \varepsilon$ is also a useful intermediate target, but only as part of a gap-closing strategy. By itself it would not imply the sharp hypothesis. What it can do is separate the problem into a certified compact region with a margin gap and a small near-threshold region that must be handled by the exact equality and KKT machinery above. In particular, the deletion reduction for a target $t < 1/6$ leaves only the closed obstruction
$$1 - 5t \leq P_{ii} \leq 5t, \qquad \lambda_{\min}(P_{\{i,j\},\{i,j\}}) \leq 4t, \qquad \lambda_{\max}(P_{\{i,j\},\{i,j\}}) \geq 1 - 4t,$$
together with
$$\lambda_{\min}(P_{TT}) \leq t, \qquad \lambda_{\max}(P_{TT}) \geq 1 - t \quad (|T| = 3).$$
Numerical minimax probes of this reduced obstruction have positive slack at $t = 1/8$ and slightly above it. The recurring boundary ansatz can be checked exactly after putting $\ell = 1 - 5t$: its active determinant equations force
$$(2q - 1)\big(3q^2 - (3 - \ell)q + \ell\big) = 0,$$
so the lower branch is
$$q(t) = \frac{2 + 5t - \sqrt{25t^2 + 80t - 8}}{6}.$$
Thus $q(1/8) = (7 - \sqrt{17})/16$, and for the stronger rational target $t = 16/125$ one gets
$$q(t) = \frac{11 - \sqrt{46}}{25} > \frac{1}{6} > \frac{16}{125}.$$

At $t = 9/70$ this branch reaches $q = 1/6$, exactly where the known seventh equality point satisfies the reduced leverage bounds. This identifies a concrete next certificate target: prove the reduced obstruction empty, first at $t = 1/8$ and then, if the same machinery survives, at $t = 16/125$.

16 Reproducibility

The accompanying repository contains the following verification scripts in `verify/`.

`v4_nesterenko.py`	exact tightness of a graphic extremal
`v6_sharpness.py`	exact sharpness certificate for the base extremal
`v9_rational_certificate.py`	six graphic-matroid sharpness certificates
`v11_seventh_exact.py`	exact reconstruction and sharpness of the seventh point
`v13_radius.py`	computed local radii at the seven points
`v16_jacobian.py`	independent Jacobian-rank isolation check
`v17_active_orbits.py`	active-set orbit counts
`v18_kkt_screen.py`	numerical active-set KKT screens
`v20_finiteness_probe.py`	actual-active-set classifier for equality hits
`v21_semialgebraic_route.py`	semialgebraic obstruction sizing
`v35_relaxed_one_ninth.py`	exact constants in the determinant relaxed proof
`v36--v41`	reduced relaxed-bound obstruction probes

The CAD cascade in Section 11 is supported by the following Sage/Singular helpers in `code/sage/`.

`probe_cad_obstruction_strata.py`	forced-coordinate dimension drops
`lexify_determinant_slice.py`	exact slices and lex bases
`classify_lex_slice_roots.py`	real-root and inequality classifier
`screen_lex_slice_minors.py`	minor-relation discovery on slices
`probe_cad_relation_open_locus.py`	component-relation checks
`probe_cad_relation_normal_forms.py`	ideal-membership checks
`reconstruct_univariate_crt.py`	modular eliminant reconstruction
`dump_determinant_gb.py`	exact determinant Groebner bases
`dump_facstd_components.py`	modular component diagnostics
`probe_s6_ansatz.py`	structured size-six ansatz probes
`verify_s6_ansatz_eliminant.py`	exact q-eliminant normal-form check
`lexify_s6_ansatz_basis.py`	exact FGLM conversion of ansatz branches
`certify_s6_ansatz_signs.py`	exact real sign checks on ansatz branches
`classify_s6_ansatz_numeric.py`	numerical classification of ansatz roots

The principal output artifacts are:

```
code/sage/out/cad_open_R0_on_I_QQ.json
code/sage/out/cad_open_R1_on_z0_QQ.json
code/sage/out/cad_open_R2_on_z0z1_QQ.json
code/sage/out/cad_obstruction_known_base_invd_QQ_z0z1z7_relz1_relz7.json
code/sage/out/lex_cad_R0_nonzero_QQmod_z0.json
code/sage/out/lex_cad_R0_nonzero_z0_eliminant_crt_6primes.json
code/sage/out/lex_cad_R0_nonzero_QQmod_z0sq_5_9_z1.json
code/sage/out/lex_cad_R0_nonzero_QQmod_z0sq_10_21_z1.json
code/sage/out/lex_cad_R0_nonzero_QQmod_z0z1sq_10_21_z2.json
code/sage/out/lex_det_mask32253_QQmod_z0.json
code/sage/out/gb_det_mask32243_QQ.json
code/sage/out/gb_det_mask31215_QQmod.json
code/sage/out/lex_det_mask32243_QQmod_u_pos_z0.json
code/sage/out/lex_det_mask31215_QQmod_u_pos_z2.json
code/sage/out/s6_ansatz_azero_QQ.json
code/sage/out/s6_ansatz_azero_fglm_QQ.json
```

```
code/sage/out/s6_ansatz_cplusd_QQ.json
code/sage/out/s6_ansatz_cplusd_q_residual_crt_6primes.json
code/sage/out/s6_ansatz_cplusd_eliminant_verify_QQ.json
code/sage/out/s6_ansatz_cplusd_fglm_QQ.json
code/sage/out/s6_ansatz_cplusd_octic_fglm_QQ.json
code/sage/out/s6_ansatz_cplusd_q3_fglm_QQ.json
code/sage/out/s6_ansatz_cplusd_q5_fglm_QQ.json
code/sage/out/s6_ansatz_cplusd_sign_cert_QQ.json
code/sage/out/s6_ansatz_structured_sign_cert_QQ.json
code/sage/out/classify_s6_ansatz_cplusd_numeric_s800.json
```

The exact claims in this paper are reproducible by running

python verify/v4_nesterenko.py,
python verify/v9_rational_certificate.py,
python verify/v11_seventh_exact.py,
python verify/v17_active_orbits.py.

The numerical KKT sweep over active sizes $10, \ldots, 13$ was run on MLCore.

job	project	region
gtz63-kkt-x79fea	test-paos	ix-m5-sm12

References

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